QUASI-COHERENT MODULES ARE A WEAK SERRE SUBCATEGORY ON THE LISSE-\'ETALE AND FLAT-FPPF SITES

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file stacks-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span cites items from Homology, Sheaves on Stacks, Descent and Cohomology of Schemes plus same-chapter lemmas, recorded in assembly.external_references with canonical tags as navigation pointers.

Allowed standard prerequisites (editorial).

- Weak Serre subcategory criterion. Stacks Project, Homology, Lemma homology-lemma-characterize-weak-serre-subcategory. Supplies the four conditions verified.
- $g_!$-$g^$* transport on the lisse-\'etale site. Stacks Project, Cohomology of Algebraic Stacks, Lemmas lemma-lisse-etale-modules, lemma-shriek-quasi-coherent, lemma-lisse-etale, lemma-quasi-coherent. Supplies $g^* g_! = \mathrm{id}$, quasi-coherence of $g_!$, exactness of g^*, and pull-back identification.
- Cokernels of quasi-coherent maps are quasi-coherent. Stacks Project, Sheaves on Stacks, Lemma stacks-sheaves-lemma-quasi-coherent-algebraic-stack. Gives the cokernel half of condition (3).
- Kernels in LQCoh^{fbc}. Stacks Project, Cohomology of Algebraic Stacks, Proposition proposition-loc-qcoh-flat-base-change. Gives the kernel half of condition (3).
- Vanishing of H^1 of quasi-coherent modules on affine schemes. Stacks Project, Descent, Propositions descent-proposition-equivalence-quasi-coherent and descent-proposition-same-cohomology-quasi-coherent; Cohomology of Schemes, Lemma coherent-lemma-quasi-coherent-affine-cohomology-zero. Assembles the $H^1 = 0$ used to lift the presentation.

Source reference. Original reference 0754: sources/stacks-chow/tags.txt, source lines 9118–9118. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0789: sources/stacks-chow/tags.txt, source lines 9229–9229. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 07B0: sources/stacks-chow/tags.txt, source lines 9325–9325. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 06WV: sources/stacks-chow/tags.txt, source lines 8829–8829. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

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Source reference. Original lemma 0788: sources/stacks-chow/tags.txt, source lines 9228–9228. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 07B1: sources/stacks-chow/tags.txt, source lines 9326–9326. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original proposition 0771: sources/stacks-chow/tags.txt, source lines 9186–9186. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0GR0: sources/stacks-chow/tags.txt, source lines 20880–20880. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 03DX: sources/stacks-chow/tags.txt, source lines 4436–4436. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 03DW: sources/stacks-chow/tags.txt, source lines 4435–4435. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01XB: sources/stacks-chow/tags.txt, source lines 2512–2512. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $\mathcal{X}$ be an algebraic stack.*

(1) *$QCoh(\mathcal{O}_{\mathcal{X}_{lisse,étale}})$ is a weak Serre subcategory of $Mod(\mathcal{O}_{\mathcal{X}_{lisse,étale}})$.*
(2) *$QCoh(\mathcal{O}_{\mathcal{X}_{flat,fppf}})$ is a weak Serre subcategory of $Mod(\mathcal{O}_{\mathcal{X}_{flat,fppf}})$.*

Proof. We will verify conditions (1), (2), (3), (4) of Homology, Lemma 0754.

Since 0 is a quasi-coherent module on any ringed site we see that (1) holds.

By definition $QCoh(\mathcal{O})$ is a strictly full subcategory $Mod(\mathcal{O})$, so (2) holds.

Let $\varphi : \mathcal{G} \to \mathcal{F}$ be a morphism of quasi-coherent modules on $\mathcal{X}_{lisse,étale}$ or $\mathcal{X}_{flat,fppf}$. We have $g^*g_!\mathcal{F} = \mathcal{F}$ and similarly for $\mathcal{G}$ and φ, see Lemma 0789. By Lemma 07B0 we see that $g_!\mathcal{F}$ and $g_!\mathcal{G}$ are quasi-coherent $\mathcal{O}_\mathcal{X}$-modules. By Sheaves on Stacks, Lemma 06WV we have that $\mathrm{Coker}(g_!\varphi)$ is a quasi-coherent module on $\mathcal{X}$ (and the cokernel in the category of quasi-coherent modules on $\mathcal{X}$). Since g^* is exact (see Lemma 0788) $g^*\mathrm{Coker}(g_!\varphi) = \mathrm{Coker}(g^*g_!\varphi) = \mathrm{Coker}(\varphi)$ is quasi-coherent too (see Lemma 07B1). By Proposition 0771 the kernel $\mathrm{Ker}(g_!\varphi)$ is in $LQCoh^{fbc}(\mathcal{O}_\mathcal{X})$. Since g^* is exact, we have $g^*\mathrm{Ker}(g_!\varphi) = \mathrm{Ker}(g^*g_!\varphi) = \mathrm{Ker}(\varphi)$. Since g^* maps objects of $LQCoh^{fbc}(\mathcal{O}_\mathcal{X})$ to quasi-coherent modules by Lemma 07B1 we conclude that $\mathrm{Ker}(\varphi)$ is quasi-coherent as well. This proves (3).

Finally, suppose that

$$0 \to \mathcal{F} \to \mathcal{E} \to \mathcal{G} \to 0$$

is an extension of $\mathcal{O}_{\mathcal{X}_{lisse,étale}}$-modules (resp. $\mathcal{O}_{\mathcal{X}_{flat,fppf}}$-modules) with $\mathcal{F}$ and $\mathcal{G}$ quasi-coherent. To prove (4) and finish the proof we have to show that $\mathcal{E}$ is quasi-coherent on $\mathcal{X}_{lisse,étale}$ (resp. $\mathcal{X}_{flat,fppf}$). Let U be an object of $\mathcal{X}_{lisse,étale}$ (resp. $\mathcal{X}_{flat,fppf}$; we think of U as a scheme smooth (resp. flat) over $\mathcal{X}$. We have to show that the restriction of $\mathcal{E}$ to $U_{lisse,étale}$ (resp. $= U_{flat,fppf}$) is quasi-coherent. Thus we may assume that $\mathcal{X} = U$ is a scheme. Because $\mathcal{G}$ is quasi-coherent on $U_{lisse,étale}$

(resp. $U_{flat,fppf}$), we may assume, after replacing U by the members of an étale (resp. fppf) covering, that $\mathcal{G}$ has a presentation

$$\bigoplus\nolimits_{j\in J} \mathcal{O} \longrightarrow \bigoplus\nolimits_{i\in I} \mathcal{O} \longrightarrow \mathcal{G} \longrightarrow 0$$

on $U_{lisse,étale}$ (resp. $U_{flat,fppf}$) where $\mathcal{O}$ is the structure sheaf on the site. We may also assume U is affine. Since $\mathcal{F}$ is quasi-coherent, we have

$$H^1(U_{lisse,étale}, \mathcal{F}) = 0, \quad \text{resp.} \quad H^1(U_{flat,fppf}, \mathcal{F}) = 0$$

Namely, $\mathcal{F}$ is the pullback of a quasi-coherent module $\mathcal{F}'$ on the big site of U (by Lemma 07B1), cohomology of $\mathcal{F}$ and $\mathcal{F}'$ agree (by Lemma 0GR0), and we know that the cohomology of $\mathcal{F}'$ on the big site of the affine scheme U is zero (to get this in the current situation you have to combine Descent, Propositions 03DX and 03DW with Cohomology of Schemes, Lemma 01XB). Thus we can lift the map $\bigoplus_{i\in I} \mathcal{O} \to \mathcal{G}$ to $\mathcal{E}$. A diagram chase shows that we obtain an exact sequence

$$\bigoplus\nolimits_{j\in J} \mathcal{O} \to \mathcal{F} \oplus \bigoplus\nolimits_{i\in I} \mathcal{O} \to \mathcal{E} \to 0$$

By (3) proved above, we conclude that $\mathcal{E}$ is quasi-coherent as desired. □